

3.2. AI Agents

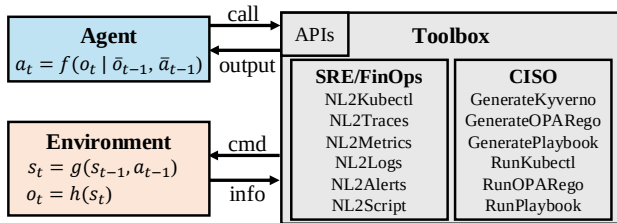


Figure 3: Agent and environment as a POMDP. Agents interact with the environment via the APIs exposed by IT-Bench’s toolbox.

In IT automation, the different personas are focused on specific desired outcome, which defines their automation

Given a scenario p instantiated by a scenario generator, an agent probes the environment and receives an observation $o_t \in \mathcal{O}$ and chooses the next action:

$$a_t = f(o_t | \bar{o}_{t-1}, \bar{a}_{t-1})$$

Here f is the agent’s decision function, which takes as input the sequence of observations up to time $t - 1$ and the sequence of actions taken up to $t - 1$.

At the beginning, o_0 may be a problematic state s_0 of the environment. Given the current state s_{t-1} and action a_{t-1} , the environment transitions to a new state:

$$s_t = g(s_{t-1}, a_{t-1})$$

The observation o_t is determined by the current state s_t and is in general a proxy for the state.